Triple Integrals: Mass and Center of Mass of a Solid

Stewart, Calculus: Early Transcendentals, Section 15.7, Exercise 39

Problem

Find the mass and the center of mass of the solid E with the given density function ρ. The solid E lies under the plane $z = 1 + x + y$ and above the region in the xy-plane that the curves $y = \sqrt{x}$, $y = 0$, and $x = 1$ bound. The density is $\rho(x, y, z) = 2$.

Symbols

- E is the three-dimensional solid, that is, the set of points (x, y, z) that satisfy the defining inequalities.
- D is the shadow of E in the xy-plane. The solid sits above D and below the plane $z = 1+x+y$.
- $dV = dz\,dy\,dx$ is the volume element. In an iterated integral it tells the reader which variable is integrated first.
- $\rho(x, y, z)$ is the density, that is, mass per unit volume. Here $\rho = 2$ is constant.
- m is the total mass, $m = \iiint_E \rho\,dV$.
- M_{yz}, M_{xz}, and M_{xy} are the first moments of mass about the planes $x = 0$, $y = 0$, and $z = 0$. For example, $M_{yz} = \iiint_E x\rho\,dV$. The subscript lists the variables that are *free* in the plane, so the distance to that plane is the remaining variable.
- $\bar{x}$, $\bar{y}$, and $\bar{z}$ are the coordinates of the center of mass. The definition is $\bar{x} = M_{yz}/m$, $\bar{y} = M_{xz}/m$, and $\bar{z} = M_{xy}/m$.

The idea of these formulas is a weighted average. The mass m is the total weight. Each first moment is the total of (coordinate) $\times$ (mass). Division of a moment by the total mass gives the average coordinate.

Part (a): mass and center of mass

Step 1: describe the solid

The base region D is bounded by the vertical line $x = 1$, the horizontal line $y = 0$, and the curve $y = \sqrt{x}$. The curve meets $y = 0$ at $x = 0$ and meets $x = 1$ at $y = 1$. Therefore

$$D = \{(x, y) : 0 \le x \le 1,\ 0 \le y \le \sqrt{x}\}. \tag{1}$$

The solid is bounded below by the plane $z = 0$ and above by the plane $z = 1 + x + y$. Hence

$$E = \{(x, y, z) : 0 \le x \le 1,\ 0 \le y \le \sqrt{x},\ 0 \le z \le 1 + x + y\}. \tag{2}$$

Figure 1 shows E: the curved top plane, the flat base D, and the three vertical walls above the boundary of D. Figure 2 shows the base region and one vertical strip.

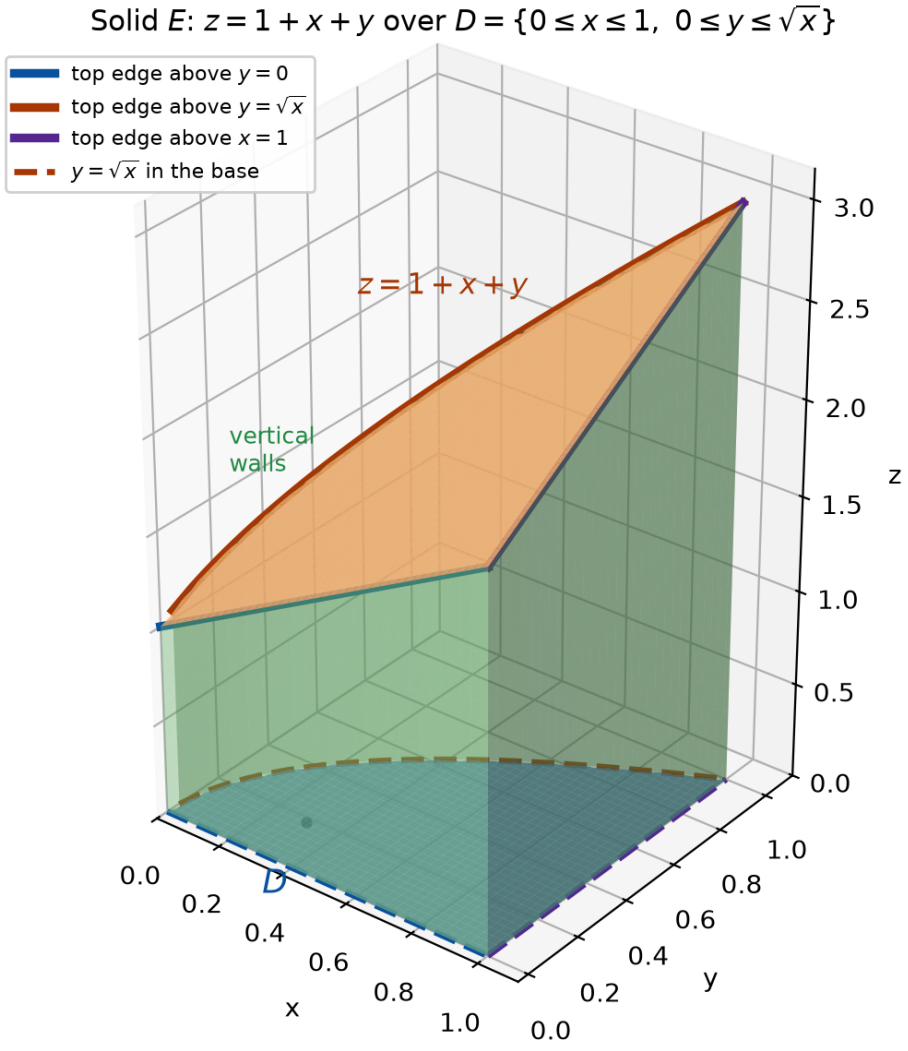


Figure 1: The solid E for Exercise 39. The orange surface is the top plane $z = 1 + x + y$. The blue surface is the base region D in the plane $z = 0$. The green surfaces are the vertical walls above the boundary of D. The thick curves are the edges of E.

Step 2: choose the order of integration

The limits above are nested. The inner limits for z depend on x and y. The middle limits for y depend on x only. The outer limits for x are constants. This order is legal because each integration removes one variable and leaves a function of the variables that are still free.

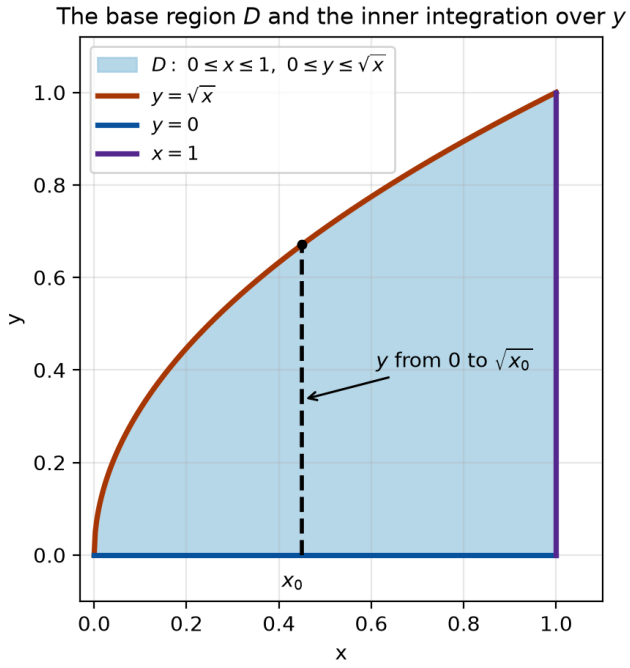


Figure 2: The base region D and a typical vertical strip at $x = x_0$. For a fixed x, the variable y runs from 0 to $\sqrt{x}$. Then x runs from 0 to 1.

The same order works for all four integrals, because the integrand of each one is a polynomial. No division by a variable occurs, and no singularity lies in E.

Step 3: compute the mass

With $\rho = 2$,

$$m = \iiint_E 2\,dV = \int_0^1 \int_0^{\sqrt{x}} \int_0^{1+x+y} 2\,dz\,dy\,dx. \tag{3}$$

The inner integral is

$$\int_0^{1+x+y} 2\,dz = 2(1 + x + y). \tag{4}$$

The middle integral is

$$\begin{aligned}\int_0^{\sqrt{x}} 2(1 + x + y)\,dy &= 2\Big[(1 + x)y + \tfrac{1}{2}y^2\Big]_0^{\sqrt{x}} \\ &= 2(1 + x)\sqrt{x} + x = 2x^{1/2} + 2x^{3/2} + x.\end{aligned} \tag{5}$$

The outer integral is

$$\begin{aligned}m &= \int_0^1 \left(2x^{1/2} + 2x^{3/2} + x\right) dx \\ &= \frac{4}{3} + \frac{4}{5} + \frac{1}{2} = \frac{40 + 24 + 15}{30} = \frac{79}{30}.\end{aligned} \tag{6}$$

Thus $m = 79/30 \approx 2.6333$. Because the density is constant, the volume is $V = m/\rho = 79/60 \approx 1.3167$.

Step 4: compute the first moments

The factor x or y or z enters the integrand. The limits do not change.

Moment about the plane $x = 0$.

$$M_{yz} = \iiint_E x\rho\, dV = \int_0^1 \int_0^{\sqrt{x}} \int_0^{1+x+y} 2x\, dz\, dy\, dx$$
$$= \int_0^1 \int_0^{\sqrt{x}} 2x(1+x+y)\, dy\, dx. \tag{7}$$

The inner integral is

$$\int_0^{\sqrt{x}} 2x(1+x+y)\, dy = 2x(1+x)\sqrt{x} + x^2 = 2x^{3/2} + 2x^{5/2} + x^2. \tag{8}$$

Then

$$M_{yz} = \int_0^1 \left(2x^{3/2} + 2x^{5/2} + x^2\right) dx$$
$$= \frac{4}{5} + \frac{4}{7} + \frac{1}{3} = \frac{84+60+35}{105} = \frac{179}{105}. \tag{9}$$

So $M_{yz} = 179/105 \approx 1.7048$.

Moment about the plane $y = 0$.

$$M_{xz} = \iiint_E y\rho\, dV = \int_0^1 \int_0^{\sqrt{x}} 2y(1+x+y)\, dy\, dx$$
$$= \int_0^1 \left(x + x^2 + \frac{2}{3}x^{3/2}\right) dx$$
$$= \frac{1}{2} + \frac{1}{3} + \frac{4}{15} = \frac{15+10+8}{30} = \frac{33}{30} = \frac{11}{10}. \tag{10}$$

So $M_{xz} = 11/10 = 1.1$.

Moment about the plane $z = 0$. Here the inner integral is

$$\int_0^{1+x+y} 2z\, dz = (1+x+y)^2. \tag{11}$$

The middle integral is

$$\int_0^{\sqrt{x}} (1+x+y)^2\, dy = \frac{(1+x+\sqrt{x})^3 - (1+x)^3}{3}$$
$$= x^{1/2} + \frac{7}{3}x^{3/2} + x^{5/2} + x + x^2. \tag{12}$$

The last line expands the difference of cubes and divides by 3. Then

$$M_{xy} = \int_0^1 \left(x^{1/2} + \frac{7}{3}x^{3/2} + x^{5/2} + x + x^2\right) dx$$
$$= \frac{2}{3} + \frac{14}{15} + \frac{2}{7} + \frac{1}{2} + \frac{1}{3} = \frac{571}{210}. \tag{13}$$

The common denominator of the last sum is 210. Thus $M_{xy} = 571/210 \approx 2.7190$.

Step 5: divide each moment by the mass

$$\bar{x} = \frac{M_{yz}}{m} = \frac{179/105}{79/30} = \frac{179}{105} \cdot \frac{30}{79} = \frac{358}{553} \approx 0.6474, \tag{14}$$

$$\bar{y} = \frac{M_{xz}}{m} = \frac{11/10}{79/30} = \frac{11}{10} \cdot \frac{30}{79} = \frac{33}{79} \approx 0.4177, \tag{15}$$

$$\bar{z} = \frac{M_{xy}}{m} = \frac{571/210}{79/30} = \frac{571}{210} \cdot \frac{30}{79} = \frac{571}{553} \approx 1.0325. \tag{16}$$

Step 6: consistency checks

First, every coordinate lies in its range: $0 < \bar{x} < 1$, $0 < \bar{y} < \sqrt{\bar{x}} = 0.8045$, and $0 < \bar{z} < 1 + \bar{x} + \bar{y} = 2.0651$. Thus the center of mass lies inside E.

Second, the moments must reproduce the mass times the coordinates:

$$\rho V \bar{x} = 2 \cdot \frac{79}{60} \cdot \frac{358}{553} = \frac{179}{105} = M_{yz}, \tag{17}$$

$$\rho V \bar{y} = 2 \cdot \frac{79}{60} \cdot \frac{33}{79} = \frac{11}{10} = M_{xz}, \tag{18}$$

$$\rho V \bar{z} = 2 \cdot \frac{79}{60} \cdot \frac{571}{553} = \frac{571}{210} = M_{xy}. \tag{19}$$

Both checks hold exactly.

Numerical confirmation

Two independent numerical methods check the exact results. The first method is a Monte Carlo average with 4×10^6 random points in the box $[0,1] \times [0,1] \times [0,3]$. The second method is a midpoint Riemann sum with $160^3 = 4.096 \times 10^6$ cells. Table 1 lists the values. The Monte Carlo error is at most 1.4×10^{-3} and the Riemann error is at most 1.3×10^{-4} in relative size.

Quantity	Exact value	Monte Carlo	Riemann sum
m	2.633333	2.634828	2.633008
M_{yz}	1.704762	1.705499	1.704809
M_{xz}	1.100000	1.099774	1.099914
M_{xy}	2.719048	2.722665	2.718885

Table 1: Exact values and the two numerical estimates for Exercise 39.

The numerical moments give the centroid (0.64729, 0.41740, 1.03334). This agrees with (358/553, 33/79, 571/553) to four decimal places.

Answer

$$\boxed{m = \frac{79}{30}, \qquad (\bar{x}, \bar{y}, \bar{z}) = \left(\frac{358}{553}, \frac{33}{79}, \frac{571}{553}\right) \approx (0.6474,\ 0.4177,\ 1.0325)} \tag{20}$$

How to solve problems of this kind

1. Draw the base region in the xy-plane first. Find where the boundary curves meet. These points give the constant outer limits.
2. Write the solid as inequalities. The order of the inequalities gives the order of integration. The innermost variable has limits that depend on the other two variables.
3. Check that the order is legal. Each limit may contain only variables that are still free at that level. The outer limits must be constants.
4. Compute the mass first. Then reuse the same limits for the moments. Only the integrand changes.
5. The center of mass is a weighted average: divide each first moment by the mass. Do not divide by the volume unless the density is constant.
6. Run a quick sanity test. The center of mass must lie inside the solid, and each moment must equal ρV times the corresponding coordinate when the density is constant.
7. Confirm with an independent numerical method. A Riemann sum or a Monte Carlo average catches sign errors and limit errors.

Summary of results

Quantity	Symbol	Value
density	ρ	2
volume	V	$79/60 \approx 1.3167$
mass	m	$79/30 \approx 2.6333$
moment about $x = 0$	M_{yz}	$179/105 \approx 1.7048$
moment about $y = 0$	M_{xz}	$11/10 = 1.1000$
moment about $z = 0$	M_{xy}	$571/210 \approx 2.7190$
centroid x	$\bar{x}$	$358/553 \approx 0.6474$
centroid y	$\bar{y}$	$33/79 \approx 0.4177$
centroid z	$\bar{z}$	$571/553 \approx 1.0325$

Table 2: All exact values for Exercise 39. Every decimal is rounded to four places.

The density is constant, so the center of mass is also the centroid of the volume. The moment test in Step 6 confirms this: each moment equals ρV times the matching coordinate. Both numerical methods in Table 1 agree with these exact values.